

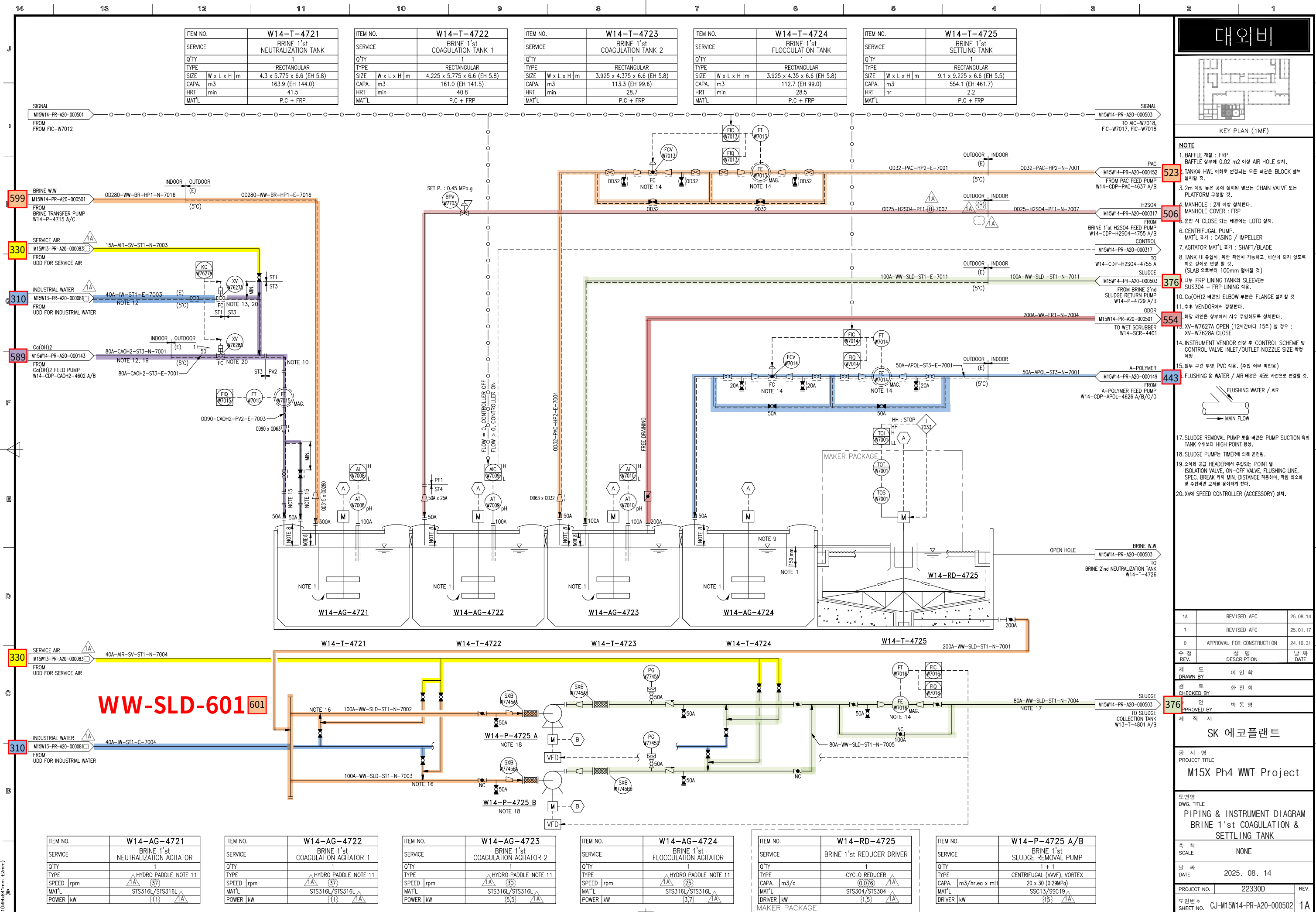
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

WW-SLD-601

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	601	1MF	100A-WW-SLD-ST1-N-7002-01	1 OF 1	0	CJ-M15W14-PR-A20-000502	1A	
2	601	1MF	100A-WW-SLD-ST1-N-7003-01	1 OF 1	1	CJ-M15W14-PR-A20-000502	1A	
3	601	1MF	200A-WW-SLD-ST1-N-7001-01	1 OF 1	0	CJ-M15W14-PR-A20-000502	1A	
4	601	1MF	40A-AIR-SV-ST1-C-7004-03	3 OF 4	0	CJ-M15W14-PR-A20-000502	1A	
5	601	1MF	40A-AIR-SV-ST1-C-7004-04	4 OF 4	0	CJ-M15W14-PR-A20-000502	1A	
6	601	1MF	40A-IW-ST1-C-7004-03	3 OF 4	0	CJ-M15W14-PR-A20-000502	1A	
7	601	1MF	40A-IW-ST1-C-7004-04	4 OF 4	0	CJ-M15W14-PR-A20-000502	1A	



대외비

KEY PLAN (1MF)

NOTE

1. Baffle 제형 : FRP

2. Baffle 상부에 0.02 m2 이상 AIR HOLE 설치.

3. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.

4. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.

5. MANHOLE : 2개 이상 설치한다.

6. 안전 시 CLOSE 되는 배관에는 LOTO 설치.

7. CENTRIFUGAL PUMP.

8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.

9. (SLAB 으로부터 100mm 떨어진 것)

10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것

11. 주후 VENDOR에서 결정한다.

12. 해당 라인은 상부에서 시수 투입하도록 설치한다.

13. XV-W7627A OPEN (12시간마다 15초) 일 경우 : XV-W7628A CLOSE

14. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.

15. 일부 구간 투명 PVC 적용. (주입 여부 확인용)

16. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.

17. SLUDGE REMOVAL PUMP 또는 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 형성.

18. SLUDGE PUMP는 TIMER에 의해 운전됨.

19. 소액의 공금 HEADER에서 주입되는 POINT 별 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 역침 최소화 및 주입배관 고제를 용이하게 한다.

20. XVM SPEED CONTROLLER (ACCESSORY) 설치.

1A

REVISED AFC

25.08.14

1

REVISED AFC

25.01.17

0

APPROVAL FOR CONSTRUCTION

24.10.31

수정

실명

날짜

REV.

DESCRIPTION

DATE

제도

이인학

DRAWN BY

검토

한진희

CHECKED BY

376

인

박동명

APPROVED BY

제작사

SK 에코플랜트

공사명

PROJECT TITLE

M15X Ph4 WWT Project

도면명

DWG. TITLE

PIPING & INSTRUMENT DIAGRAM

BRINE 1'st COAGULATION & SETTLING TANK

속척

SCALE

NONE

날짜

DATE

2025. 08. 14

PROJECT NO.

22330D

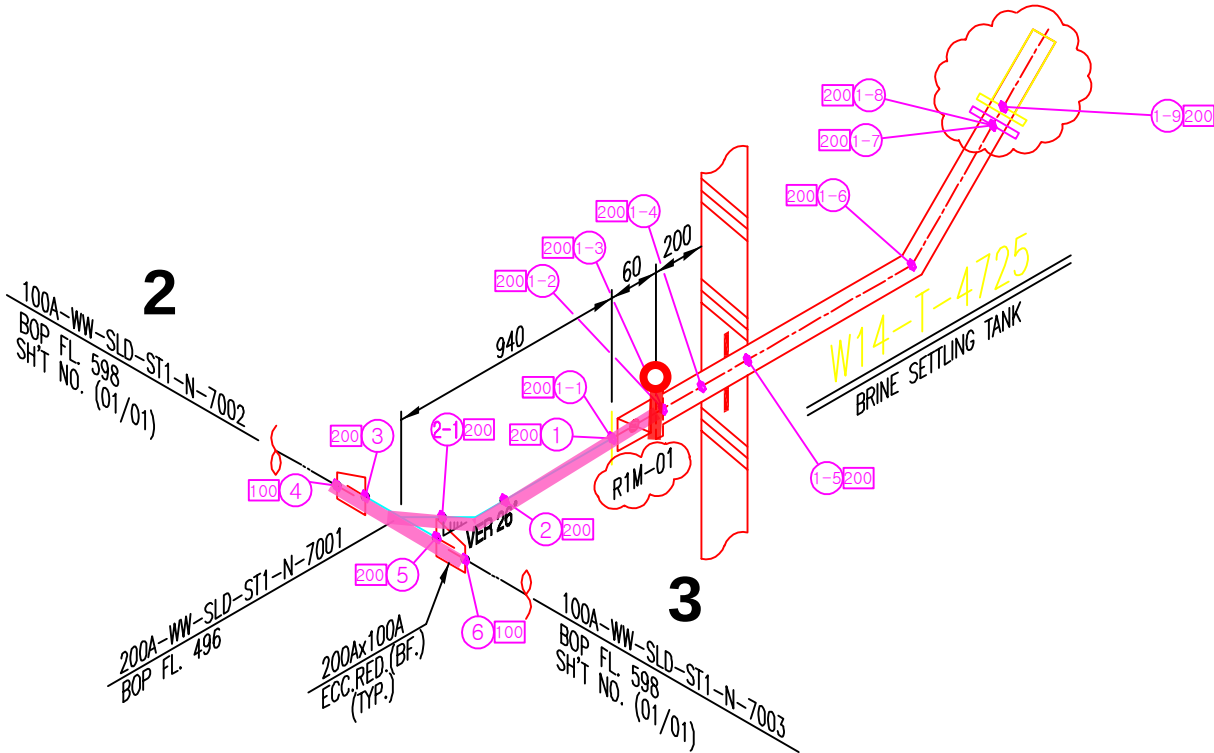
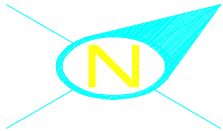
REVISION

1A

도면번호

SHEET NO.

CJ-M15W14-PR-A20-000502



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								
	THICK.	—	DESIGN	6.4	60	UNIT	R1MF								
			OPERATING	4	AMB	REF. PLAN DWG.	CJ—M15W14—PI—A30—010018—C								
						REF. P. & ID	CJ—M15W14—PR—A20—000502	△	REV. NO.	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	
									DATE		DESCRIPTION	BY	CHK	APP'D	



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	200A	0.9M	PIPE, STS 304TP, ERW, BE – SCH 10S
T	200A	1EA	TEE EQUAL, STS304, ERW, BE – SCH 10S
RE	200Ax100A	2EA	REDUCER ECC, STS304, ERW, BE – SCH 10S
VF	200A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY—GCD450 / TRIM—STS304 / EPDM—LINED, GEAR
F	200A	1EA	FLG SO, KS 10K, RF, STS304
BL	200A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)
E26	150A	1EA	ELBOW 23 DEG SR, STS304, ERW, BE – SCH 10S

ST1			
P	200A	5.8M	PIPE, STS 304TP, ERW, BE – SCH 10S
F1	200A	2EA	FLG SO, KS 10K, FF, STS304
STOP	200A	1EA	ANCHOR PLATE
G1	200A	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	200A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

FR1			
P2	200A	0.7M	PIPE, FRP(MANUFACTURER STD), PE/BUTT—JOINT, 4.5T
F2	200A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT—JOINT

CLIENT



1

CONTRACTOR		SK ecoplant		PROJECT NAME	
COMMODITY		WW—SLD		JOB NO.	
LINE NO.		200A—WW—SLD—ST1—N—7001		22330D	
				SH'T NO.	REV.
				01 / 01	0



2

CLIENT  2

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WW—SLD		22330D	
LINE NO.	SH'T NO.	REV.	
100A-WW-SLD-ST1-N-7002	01 / 01	0	

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	—	DESIGN	6.4	60	UNIT R1MF						
			OPERATING	4	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-010018-C						
						REF. P & ID CJ-M15W14-PR-A20-000502		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP



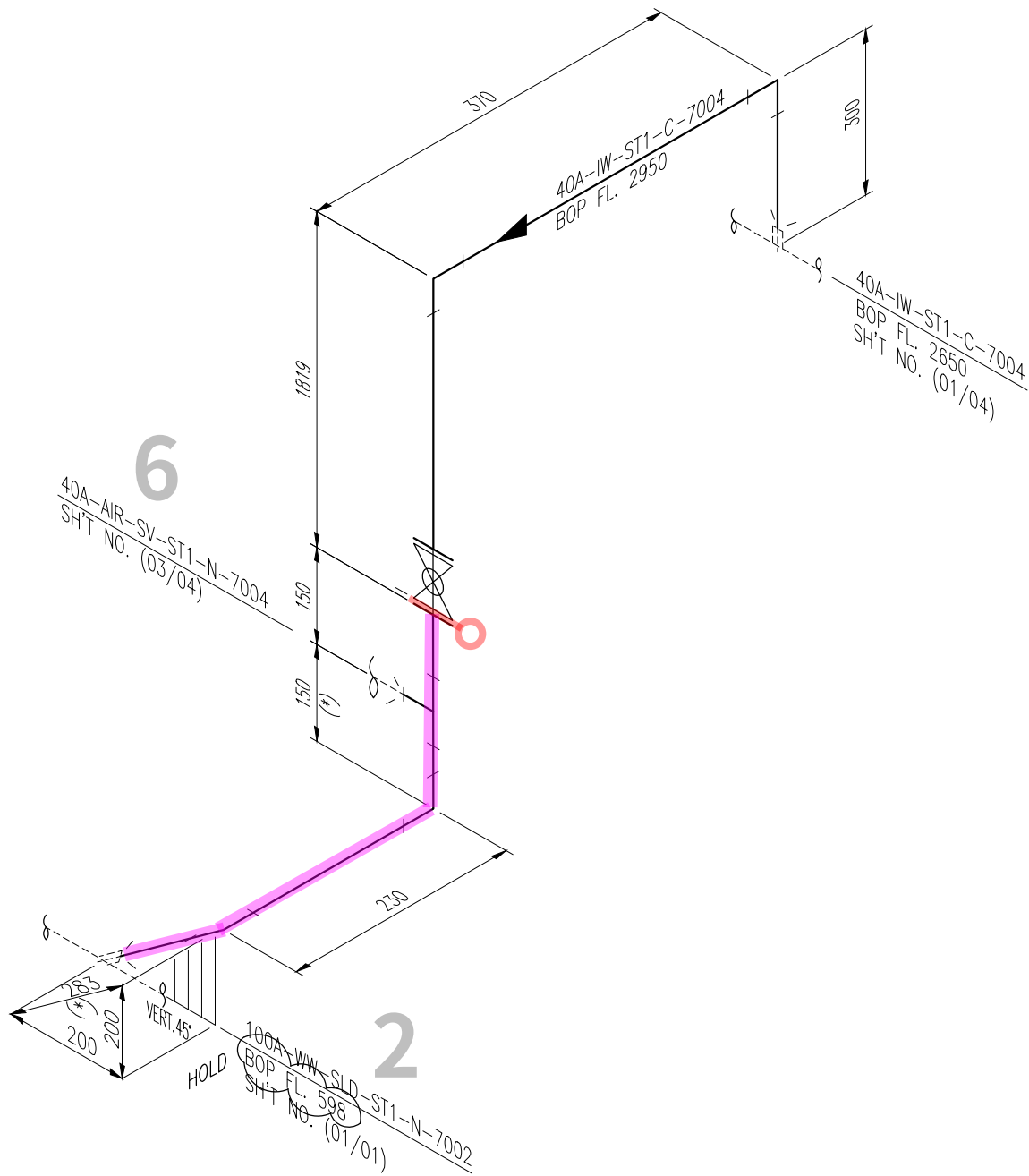
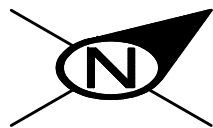
CLIENT



3

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	6.4	60	UNIT R1MF								M15X Ph4 Project		
			OPERATING	4	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-010018-C		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	WW-SLD	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000502		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. 100A-WW-SLD-ST1-N-7003		22330D	
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			SH'T NO. 01 / 01	REV. 1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	13.5	60	UNIT	R1MF									SK ecoplant	M15X Ph4 Project
			OPERATING	10	AMB	REF. PLAN DWG.										COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-010018-C										IW	22330D
						REF. P & ID										LINE NO.	SH'T NO. REV.
						CJ-M15W14-PR-A20-000502										40A-IW-ST1-C-7004	03 / 04 0

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[illegible]

<NOTE>
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* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB1.1/VB2/VB2.0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
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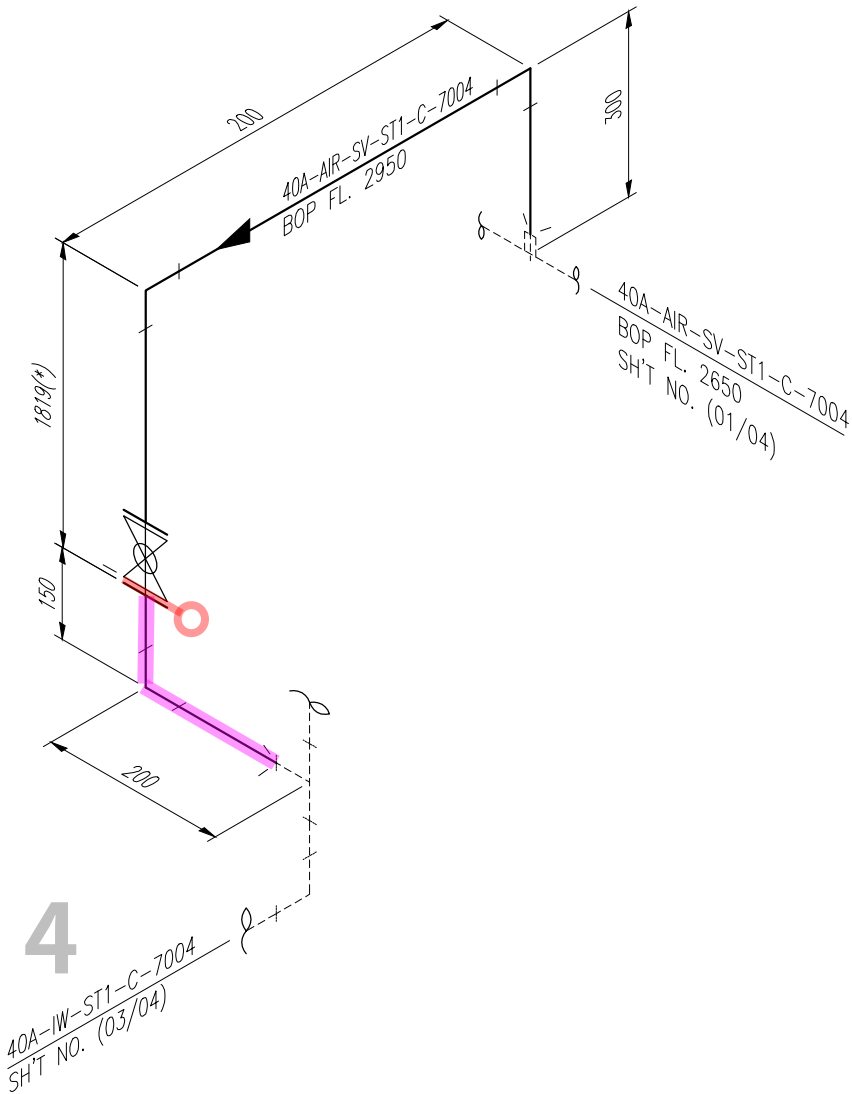
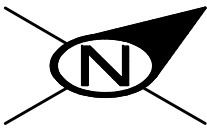
CLIENT



5

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT	R1MF								M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG.								COMMODITY	IW	JOB NO.
						CJ-M15W14-PI-A30-010018-C		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	40A-IW-ST1-C-7004	SH'T NO.	REV.
					REF. P & ID	REV. NO.										
						CJ-M15W14-PR-A20-000502									04 / 04	0

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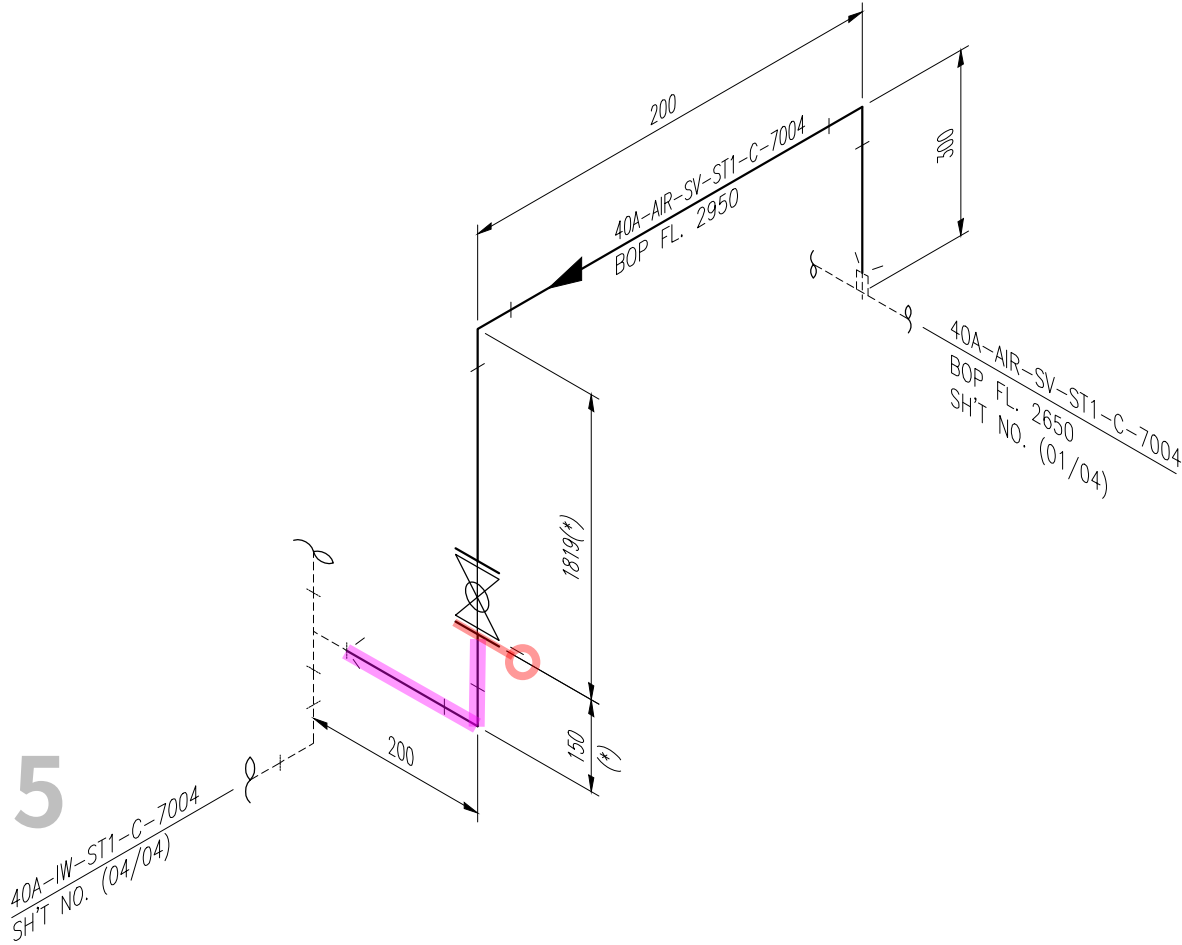
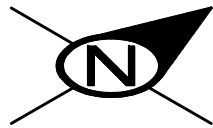


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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14								CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	10	60	UNIT R1MF								SK ecoplant	M15X Ph4 Project
			OPERATING	7.5	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-010017-D								COMMODITY	JOB NO. 22330D
						REF. P & ID CJ-M15W14-PR-A20-000502								LINE NO.	SH'T NO. REV.
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		40A-AIR-SV-ST1-C-7004	03 / 04 0

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* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	10	60	UNIT	R1MF									SK ecoplant	M15X Ph4 Project
			OPERATING	7.5	AMB	REF. PLAN DWG.										COMMODITY	AIR-SV
						CJ-M15W14-PI-A30-010017-D										LINE NO.	40A-AIR-SV-ST1-C-7004
						REF. P & ID										SH'T NO.	04 / 04
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경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.